

GridWatch: Predicting Energy Burden

Across Cities — Progress Report

Abstract

GridWatch is a data-driven system designed to predict energy burden across cities by integrating demographic, weather, and electricity pricing data. The project delivers an end-to-end pipeline spanning data ingestion, preprocessing, predictive modeling, explainability via SHAP, and a generative AI reasoning layer. This report summarizes the work completed to date, key challenges encountered, and the plan for the remaining development phase.

INTRODUCTION

Energy burden — the share of household income consumed by energy costs — disproportionately affects low-income and climate-vulnerable communities. GridWatch addresses this by building a reproducible, multi-source pipeline that identifies high-risk regions and surfaces actionable insights for policymakers, researchers, and energy planners. A functional prototype has been implemented covering all major system components.

WORK COMPLETED

Data Collection

Multiple public datasets were integrated, including U.S. Census demographic records, historical weather data, electricity pricing time series, and energy burden indicators. Modular ingestion scripts automate retrieval and ensure reproducibility. Key features collected include median household income, population density, housing characteristics, temperature and seasonal variation, and utility pricing. Raw and processed datasets are maintained in separate structured directories to support traceability.

Data Cleaning & Preprocessing

Preprocessing scripts standardize each dataset prior to integration. Steps include missing value imputation, column name normalization, geographic identifier alignment, data type conversion, and timestamp formatting. Feature scaling was applied to ensure compatibility across sources. Derived features — such as energy burden indicators and aggregated seasonal weather statistics — were computed at this stage to enrich the eventual training dataset.

Data Merging & Feature Engineering

Cleaned datasets were merged into a unified training corpus by aligning city-level geographic keys across sources. Feature engineering combined demographic indicators, weather statistics, and electricity pricing variables, with additional derived features constructed to capture economic-environmental interactions. The merged dataset is persisted in the processed data directory, decoupling model training from data ingestion and improving experimental reproducibility.

Model Training

A machine learning pipeline was implemented to predict energy burden levels. The training script loads the processed dataset, partitions it into training and test splits, and fits a predictive model. Preliminary results confirm that the pipeline runs end-to-end and produces meaningful predictions. The model successfully identifies correlations between income levels, temperature variability, and electricity pricing, demonstrating the ability to distinguish high-burden from low-burden regions.

Explainability Integration

SHAP-based feature importance analysis was integrated to ensure model interpretability. This module quantifies each variable's contribution to individual predictions. Early results indicate that median income, electricity price, and extreme temperature range are among the most influential predictors — findings that are consistent with domain expectations and support stakeholder trust in the model's outputs.

Generative AI Reasoning

A generative AI reasoning layer was implemented to translate numerical model outputs into coherent, human-readable explanations. This component summarizes predictions and articulates the key drivers behind them — for example, explaining why a specific city is flagged as high-risk. This capability significantly improves usability for non-technical stakeholders such as city planners and policy analysts.

Working Prototype

An application entry point orchestrates the full pipeline: loading processed data, running model inference, and producing reasoning output. The prototype demonstrates a functional end-to-end workflow from raw data ingestion through to interpretable, actionable results.

CHALLENGES ENCOUNTERED

Geographic Identifier Alignment. Different data sources employed inconsistent naming conventions for cities and regions. This was resolved by implementing a standardized identifier mapping layer during preprocessing.

Missing Data. Census and weather datasets contained gaps that required careful imputation and filtering strategies to avoid introducing systematic bias into the training set.

Feature Scaling. Merging demographic and weather features with disparate numerical ranges introduced scaling issues, which were mitigated through normalization techniques applied during preprocessing.

Pipeline Modularity. Organizing multiple interdependent scripts into a coherent, reproducible pipeline required significant repository restructuring. The resulting modular architecture has improved both maintainability and clarity.

REVISED PLAN

The next development phase focuses on model refinement and output quality. Planned tasks include hyperparameter tuning, cross-validation, and a systematic comparison of evaluation metrics. Additional feature engineering will be explored to further improve predictive accuracy, and the generative AI prompts will be refined to yield clearer, more actionable explanations.

Optional enhancements under consideration include building an interactive visualization dashboard and preparing a deployment configuration. The final weeks will be dedicated to documentation, final report preparation, and the development of presentation materials.

CONCLUSION

A working prototype of GridWatch has been successfully implemented, encompassing data ingestion, preprocessing, model training, SHAP-based explainability, and generative AI reasoning. The system demonstrates full end-to-end functionality and produces interpretable, stakeholder-ready insights. Remaining efforts are focused on strengthening model performance, refining natural-language explanations, and delivering final project deliverables.